

# MONICA R

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## ABOUT

Enthusiastic Software Developer with a strong foundation in programming and problem-solving. Skilled in Java, Python and basic software development concepts. Eager to apply technical knowledge to build efficient applications and grow in a challenging development environment.

## Education

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| <b>IFET College Of Engineering Villupuram, BE in Electronics and Communication Engineering</b> | Nov 2022 – May 2026 |
| <ul style="list-style-type: none"><li>CGPA: 8.17/10</li></ul>                                  |                     |
| <b>Amalorpavam Higher Secondary School, Pondicherry.</b>                                       | Jun 2021 – May 2022 |
| <ul style="list-style-type: none"><li>HSC Percentage: 80</li></ul>                             |                     |

## Experience

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| <b>INTERN – Web Developer</b> – <i>LeMenz Technologies Pvt. Ltd., Pondicherry</i>                                                                                                                                                                                                                                         | 2024 |
| <ul style="list-style-type: none"><li>Gained hands-on experience in frontend and backend development through real-time projects.</li><li>Improved skills in UI design, API integration, and web/application development frameworks.</li></ul>                                                                             |      |
| <b>INTERN – Python Full Stack Developer</b> – <i>VEI Technologies Pvt. Ltd., Tindivanam</i>                                                                                                                                                                                                                               | 2025 |
| <ul style="list-style-type: none"><li>Strengthened backend development skills using Python and frameworks such as Django and Flask.</li><li>Acquired practical experience in database management, SQL queries, and frontend–backend integration.</li></ul>                                                                |      |
| <b>INPLANT TRAINING - Embedded Systems &amp; IoT Trainee</b> – <i>TVS Training Services Ltd., Chennai</i>                                                                                                                                                                                                                 | 2025 |
| <ul style="list-style-type: none"><li>Worked with microcontrollers, sensors, and electronic circuit design in real-world scenarios.</li><li>Developed skills in Embedded C, firmware testing, and hardware debugging.</li><li>Learned PCB fundamentals, wiring techniques, and embedded system problem-solving.</li></ul> |      |

## Publications

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| <b>1. Arduino-Based Alcohol Detection Device: Enhancing Safety in Vehicle Operation through Sensor Technology,</b> | March 2025 |
| <i>Asian Journal of Applied Science and Technology, Vol. 8, Issue 1</i>                                            |            |
| <b>2. A Novel and Reliable SOS Alert Band System for Women Safety,</b>                                             | March 2024 |
| <i>Asian Journal of Applied Science and Technology, Vol. 9, Issue 1</i>                                            |            |

## Projects

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| <b>ARDUINO BASED TOLL TAX SYSTEM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2023 |
| <ul style="list-style-type: none"><li>Developed an automated toll collection system using Arduino with sensors (IR/Rfid) for vehicle detection and identification.</li><li>Implemented real-time toll calculation and automatic transaction handling to ensure fast and accurate processing</li><li>Integrated LCD display and servo motor barrier mechanism to streamline traffic flow and enhance operational efficiency</li><li>Designed an embedded system architecture combining <b>sensor integration, microcontroller programming, and automation control</b> for smart transportation applications.</li><li>Tools Used: Arduino Idle, Microcontroller</li></ul> |      |

## **VIRO SHIELD - INTELLIGENT DOOR AUTOMATION SYSTEM**

2024

- Initiated and led the creation of an automated entry control system integrating temperature screening sensors to enhance security and health safety.
- Managed hardware and software development efforts ensuring project success
- Enhanced safety and energy efficiency by integrating alert signals and auto-close functionality based on sensor feedback.
- Tools Used: IOT, Arduino Idle

## **SECURESHE - WEARABLE SAFETY DEVICE**

2025

- Conceptualized and Led the design of an offline GSM-based wearable safety device with nerve sensors on the foot thumb for discreet panic detection. The device triggers automatic SOS alerts using biometric and environmental sensors without mobile dependency.
- Enhanced user safety with quick-response mechanisms, LED/Vibration indicators, and reliable real-time activation.

## **CAPACITY ANALYSIS OF MIMO-OFDM WIRELESS COMMUNICATION SYSTEM UNDER RAYLEIGH FADING CHANNEL USING ADAPTIVE SNR OPTIMIZATION**

2026

- Developed a **MIMO-OFDM Wireless Communication System** under **Rayleigh Fading Channel** for reliable and high-speed wireless data transmission.
- Designed and simulated a **Microstrip Patch Antenna** using Keysight ADS to improve **signal gain, bandwidth,** and communication efficiency.
- Implemented **Adaptive SNR Optimization** to enhance **channel capacity, spectral efficiency,** and overall wireless network performance.
- Analyzed key communication parameters including **BER, SNR, throughput,** and **frequency response** for system performance evaluation.
- Developed an interactive dashboard using Streamlit for real-time visualization and monitoring of wireless communication performance metrics, supporting next-generation **5G communication systems.**
- Enhanced user safety with quick-response mechanisms, LED/Vibration indicators, and reliable real-time activation.
- Tools used: ADS software, VS code, Streamlit run app.py

## **Technologies**

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**Languages:** Java, Python, C , C + + , SQL, JavaScript , HTML, CSS

**Technologies:** .NET, Node.js

## **Certifications**

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1. **Java Full Stack Certification – Wipro**
2. **Drone Technology Certification – Atal Incubation Center, Pondicherry**
3. **CCNA – Cisco Certified Network Associate**